3 b) For the given set of operations, design a 4-bit universal shift register. 4

Mode	operation
00	Hold
01	Shift Left
10	Shift Right
11	Complement the content

4 b) Construct a Johnson counter with 10 timing signals. Compare Johnson and Ring counters. 5

1. With a neat diagram, explain the operation of NOR Logic (ECL). Also mention the advantages and disadvantages of Logic (ECL). Also mention the advantages and disadvantages of

2 Design an arithmetic and logic circuit with three selection variables $S_2S_1S_0$ that generate the following operations. Draw the logic diagram of one typical stage.

Arithmetic Operations					Logical Operations			
S_2	S_1	S_0	$C_{in}=0$	$C_{in}=1$	S_2	S_1	S_0	$C_{in}=X$
0	0	0	F=A	F=A+1	1	0	0	OR
0	0	1	F=A+B	F=A+B+1	1	0	1	XOR
0	1	0	F=A-B-1	F=A-B	1	1	0	AND
0	1	1	F=A-1	F=A	1	1	1	Complement

4 b) What is a scratch pad memory? Explain the organization of a processor unit using scratch pad memory with the help of a neat diagram. 6

8 a Design a 4-bit synchronous binary up/down counter using T flip flops. 06

b Draw the logic diagram of a four bit register with four D flip-flops and 4×1 multiplexers with mode selection inputs S_1 and S_0. The register operates according to the function table (8b). 06

Table 8(b)

S_1	S_0	*Register operation*
0	0	*No change*
0	1	*Complement the four outputs*
1	0	*clear register*
1	1	*load parallel data*

) a Draw the block diagram of a processor unit that uses a scratch pad memory. Illustrate its usage to perform the operation $R_1 \leftarrow R_2 + R_3$. bits in a status register? How are these bits set or

a	Show the realization of a mod-90 counter using 7490 decade counter.	06
b	Design a 4 bit bidirectional shift register with parallel load. Write the function table showing different operations.	10

Show the realization of a synchronous 4- bit binary up down counter using JK FFs. Show the timing diagrams for up counting operation. Show the realization of a 10 state Johnson counter along with the state diagram.

10 a Describe the bus organization of processor registers and ALU with diagrams. 06

b Design an arithmetic circuit, with two selection variables S_1 and S_0, that generates the following arithmetic operations. Draw the logic diagram of one typical stage.

S_1	S_0	$C_{in} = 0$	$C_{in} = 1$
0	0	$F = A + B$	$F = A + B + 1$
0	1	$F = A$	$F = A + 1$
1	0	$F = \overline{B}$	$F = \overline{B} + 1$
1	1	$F = A + \overline{B}$	$F = A + \overline{B} + 1$

10

OR

11 a Design an arithmetic and logic circuit, with three selection variables S_2, S_1 and S_0, that generates the following operations. Draw the logic diagram of one typical stage.

Arithmetic Operations:					*Logical Operations:*				
S_2	S_1	S_0	$C_{in} = 0$	$C_{in} = 1$	S_2	S_1	S_0	$C_{in} = X$	
0	0	0	$F = A$	$F = A + 1$	1	0	0	$F = A \vee B$	OR
0	0	1	$F = A + B$	$f = A + B + 1$	1	0	1	$F = A \oplus B$	$EX - OR$
0	1	0	$F = A - B - 1$	$F = A - B$	1	1	0	$F = A \wedge B$	AND
0	1	1	$F = A - 1$	$F = A$	1	1	1	$F = \overline{A}$	Complement

1

10	a	Describe with a suitable diagram the bus organization of a processor register with ALU.
	b	Design an arithmetic circuit with one selection variable S and two data inputs A and B. When $S = 1$, circuit should perform the addition operation $F = A + B$. When $S = 0$, the circuit should perform the increment operation $F = A + 1$. Draw the logic diagram of a typical stage.
	c	Write a descriptive note on status register.

OR

11	a	Design an arithmetic circuit with two selection variables s_1 and s_0, which generates the arithmetic operation shown in table 1. Modify the arithmetic circuit into an ALU with mode select variable S_2.When $S_2 = 0$, the ALU is identical to the arithmetic circuit, when $S_2 = 1$, it generates the logic functions according to table 2. Draw the logic diagram of two typical stages.

Table 1: Arithmetic operations

S_2	S_1	S_0	$C_{in} = 0$	$C_{in} = 1$
0	0	0	$F = A$	$F = A + 1$
0	0	1	$F = A + B$	$F = A + B + 1$
0	1	0	$F = A - B - 1$	$F = A - B$
0	1	1	$F = A - 1$	$F = A$

Table 2: Logical operations

S_2	S_1	S_0	$C_{in} = 0$
1	0	0	$F = A \vee B$
1	0	1	$F = A \oplus B$
1	1	0	$F = A \wedge B$
1	1	1	$F = A$

b) Show the realization of a ring counter with 6 states.

a)	Design a 4 bit Universal shift register and draw the circuit with the given mode of operation table.	10

S_1	S_0	Operation
0	0	Parallel
0	1	Shift Right
1	0	Shift Left
1	1	Inhibit Clock